

HAT-P-32b

R-0582

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_221228 · created 2026-07-24T18:29:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459941.6616 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.037
Transit depth (Rp/Rs)^2	4.7e-06 +/- 0.0016 [%]
Orbital Inclination (inc)	85.31 +/- 1.71
Airmass coefficient 1 (a1)	1.38 +/- 0.47
Airmass coefficient 2 (a2)	-0.23 +/- 0.17
Scatter in the residuals of the lightcurve fit is	34.04 %
Best Comparison Star	#2 - [629, 251]
Optimal Aperture	1.39
Optimal Annulus	8.34
Transit Duration (day)	0.1077 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.037 vs 0.1489 (deviation 1.06σ)
Duration fit vs published	0.1077 d vs 0.1304 d (deviation 0.82σ)
Mid-time O-C	-9.1 min
Depth SNR	0.0
Residual scatter	34.04 %

Light curve

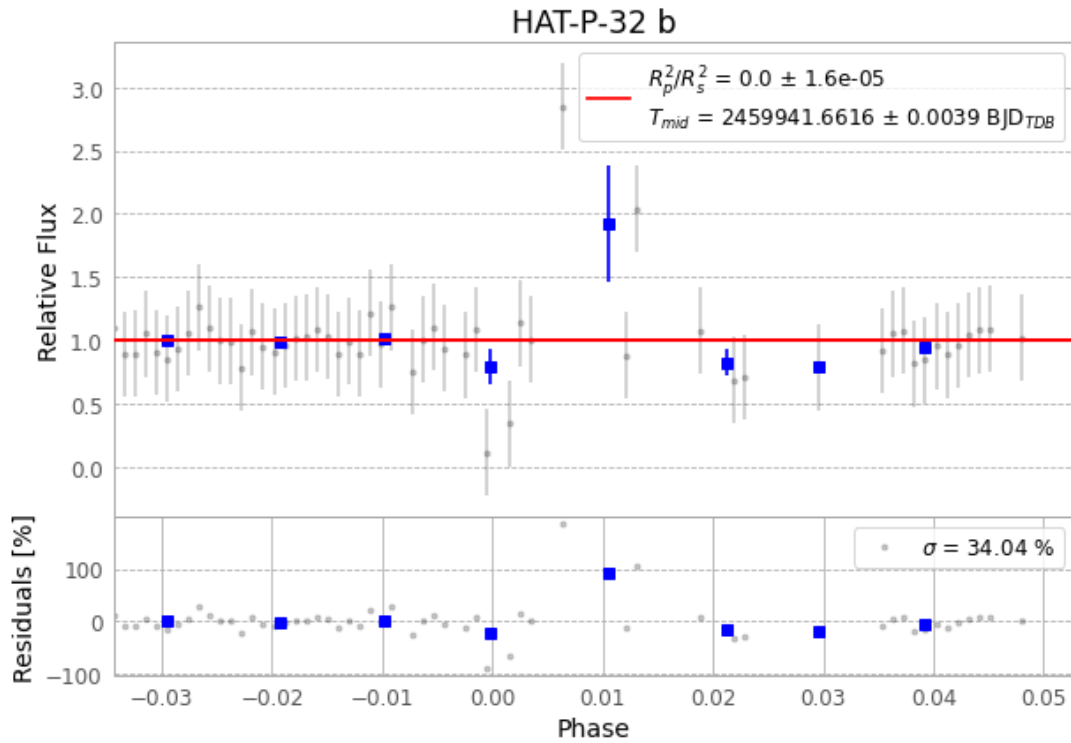


Figure 1 — FinalLightCurve_HAT-P-32 b_2022-12-27.png (SHA-256 fc577367ac9876d1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [502, 174], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 94dbe1cebf266a38...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

91 FITS frames (60 MB), HATP-32221228020016.FITS → HATP-32221228063016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-221228.log (56c9cbd0eee1...), FinalLightCurve_HAT-P-32 b_2022-12-27.png (fc577367ac98...), FinalLightCurve_HAT-P-32 b_2022-12-27.csv (1a0269add19b...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 18:29Z.